Intro to Data Science - Lab 6

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Week 6 - Using ggplot to Build Complex Data Displays

```
# Enter your name here: Sofia Quintero
```

Please include nice comments.

Instructions:

Run the necessary code on your own instance of R-Studio.

Attribution statement: (choose only one and delete the rest)

```
# 1. I did this lab assignment by myself, with help from the book and the professor.
```

Creating graphical displays of data is an essential skill for all data scientists because so much of what data scientists do involves communicating with other people about data. The **ggplot2** package developed by data scientist Hadley Wickham (https://en.wikipedia.org/wiki/Hadley_Wickham) provides excellent power and flexibility for graphically displaying data. Whole books have been written about **ggplot2** (e.g.: https://www.springer.com/gp/book/9780387981413), so we will only be able to scratch the surface, but we will master the basic grammar that you need in order to use this
package. For this lab we will also use a dataset that comes delivered in R, called ** economics **. It contains 574 snapshots of U.S. economic conditions between 1967 and 2014.
Here are two lines of starter code:

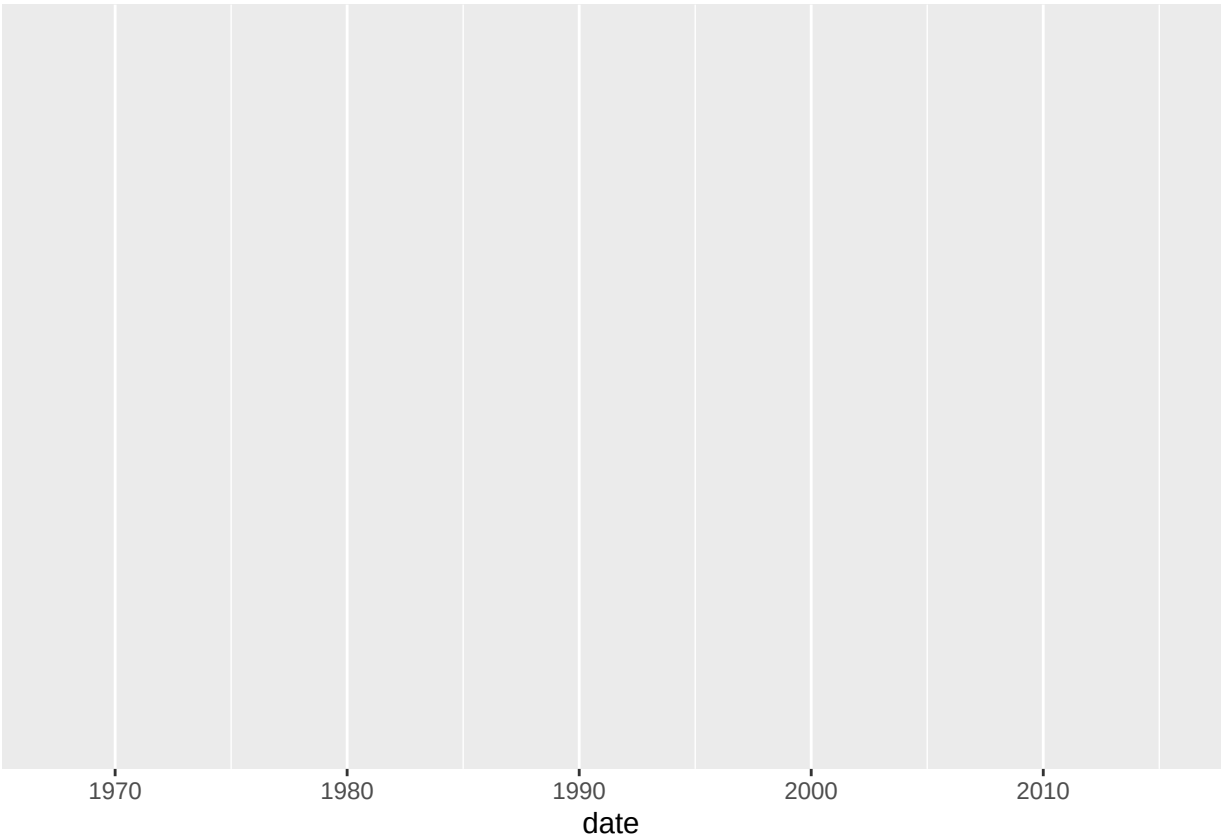
```
MyPlot <- ggplot(economics, aes(x=date))
myPlot <- myPlot + geom_line(aes(y=psavert))
```

1. Run these two lines of code below. What happens? How do you actually invoke the plot (i.e., how do you get it to draw in the plot window)?

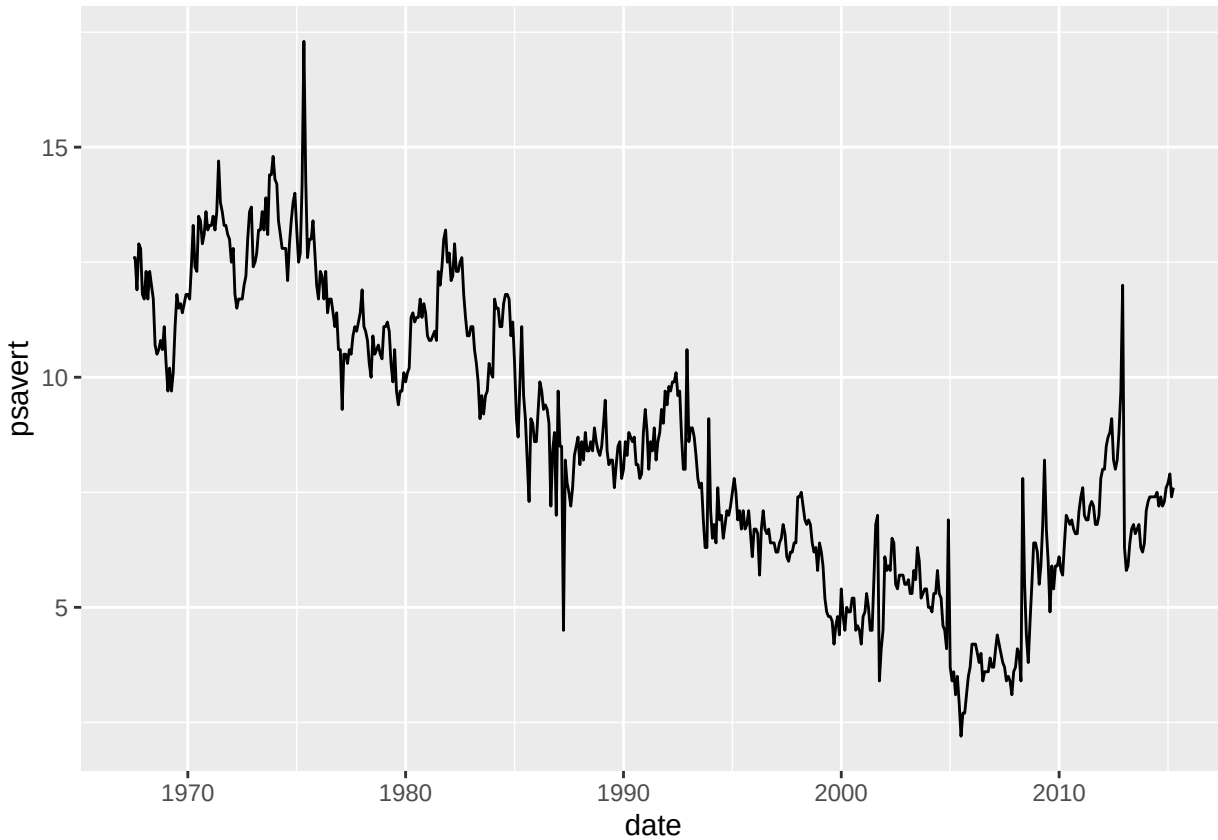
```
install.packages('ggplot2', repos = "http://cran.us.r-project.org")
```

```
##
## The downloaded binary packages are in
##  /var/folders/rm/s4vv52kd55g34xzsxxqn9p2c0000gn/T//RtmpFBzpg2/downloaded_packages
```

```
library(ggplot2)
MyPlot <- ggplot(economics, aes(x=date))
myPlot <- MyPlot + geom_line(aes(y=psavert))
MyPlot
```




```
myPlot
```




```
#MyPlot creates a plot but no data is drawn, possibly due to the absence of a y-value
#to plot with
#myPlot has a y-value, therefore the plot is drawn with geom_lines() connecting the data
```

2. Run `help("economics")` to find out the meaning of the **psavert** variable.

```
help("economics")
#psavert is the personal savings rate
```

3. Examine the plot to estimate when the personal savings rate reached its maximum value. Also examine the plot to estimate when the personal savings rate reached its minimum value.

```
#psavert reached its maximum value in 1975
#psavert reached its minimum value around 2005
```

4. Use **which.max( )** and **which.min( )** to verify your guesses from problem 3.

```
#Max:
economics$date[which.max(economics$psavert)]
```

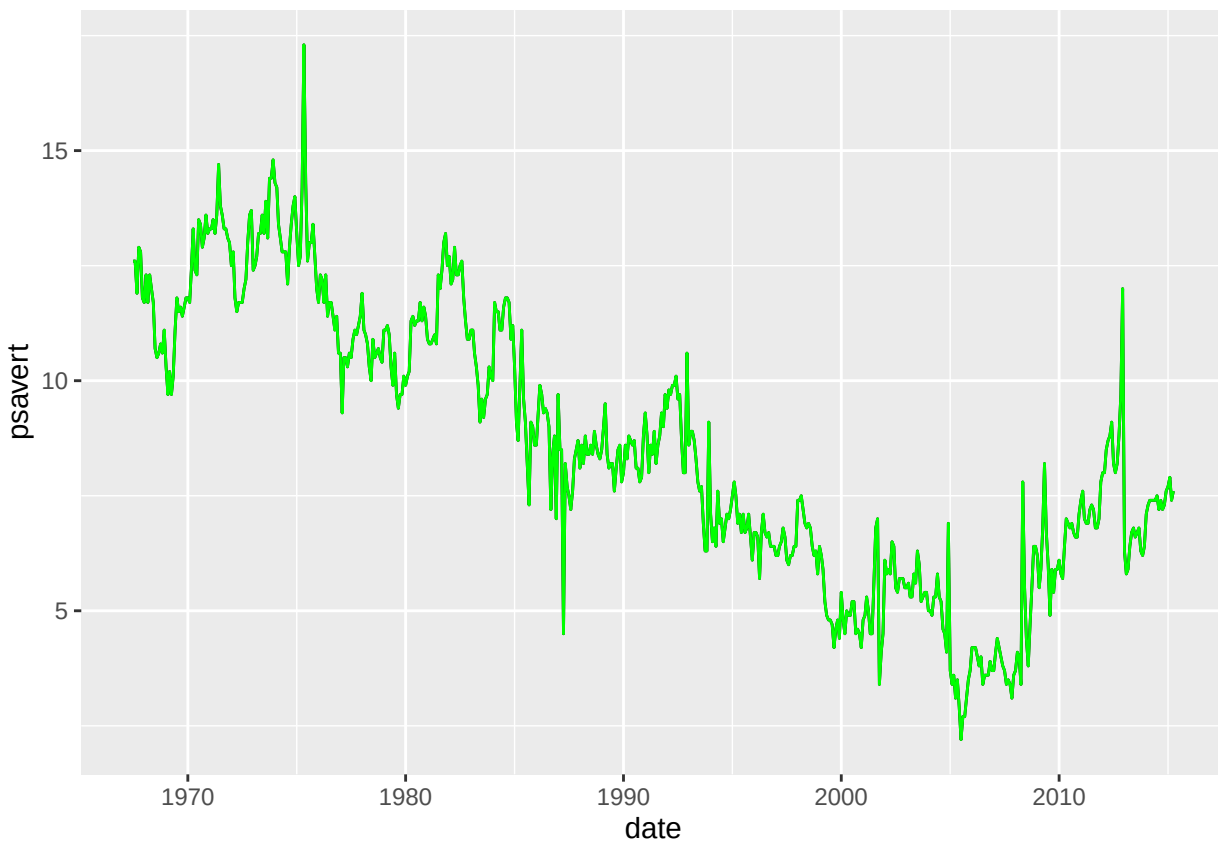
```
## [1] "1975-05-01"
```

```r
#Min:
economics$date[which.min(economics$psavert)]
```

```
## [1] "2005-07-01"
```

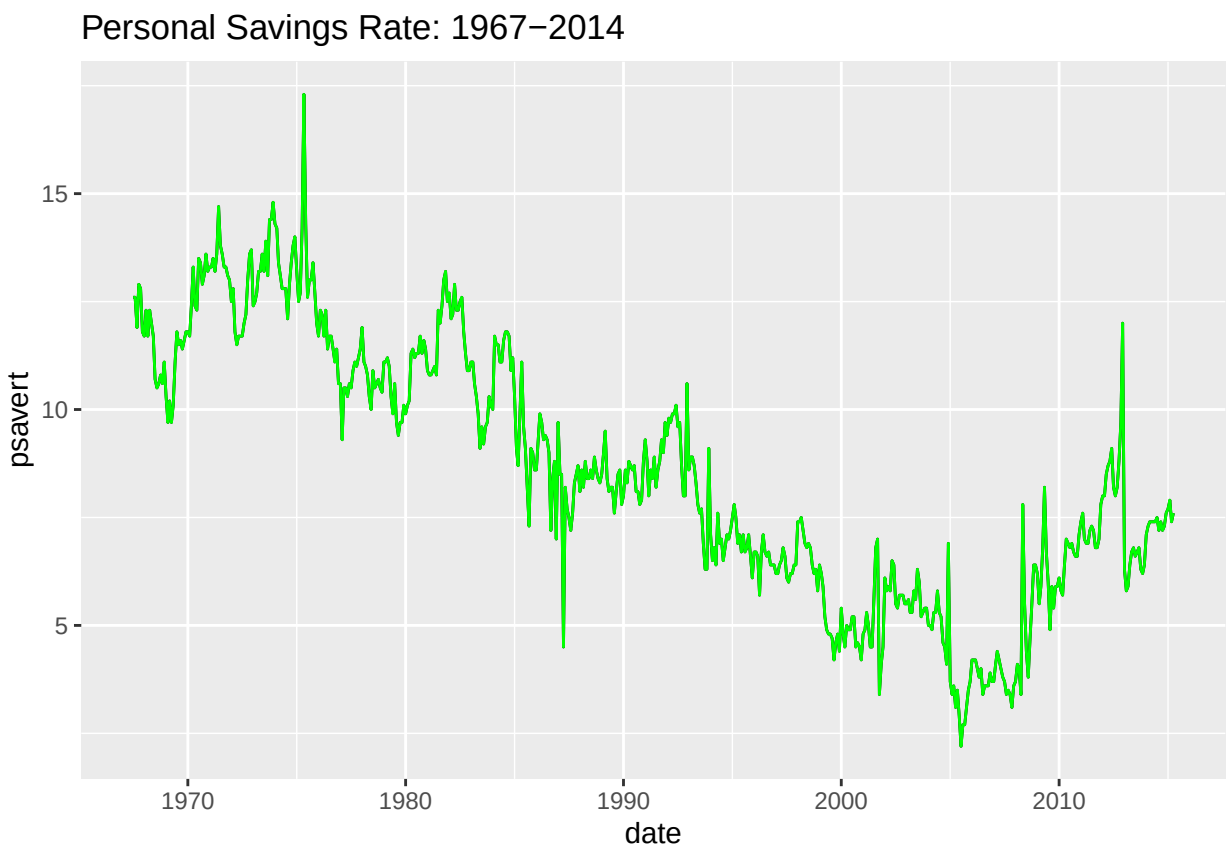
5. Change the color of the plot line to green. **Hint:** Changing a line to a constant color happens in the specification of the **geometry**.

```r
myPlot <- myPlot + geom_line(aes(y=psavert), colour = "green")
myPlot
```



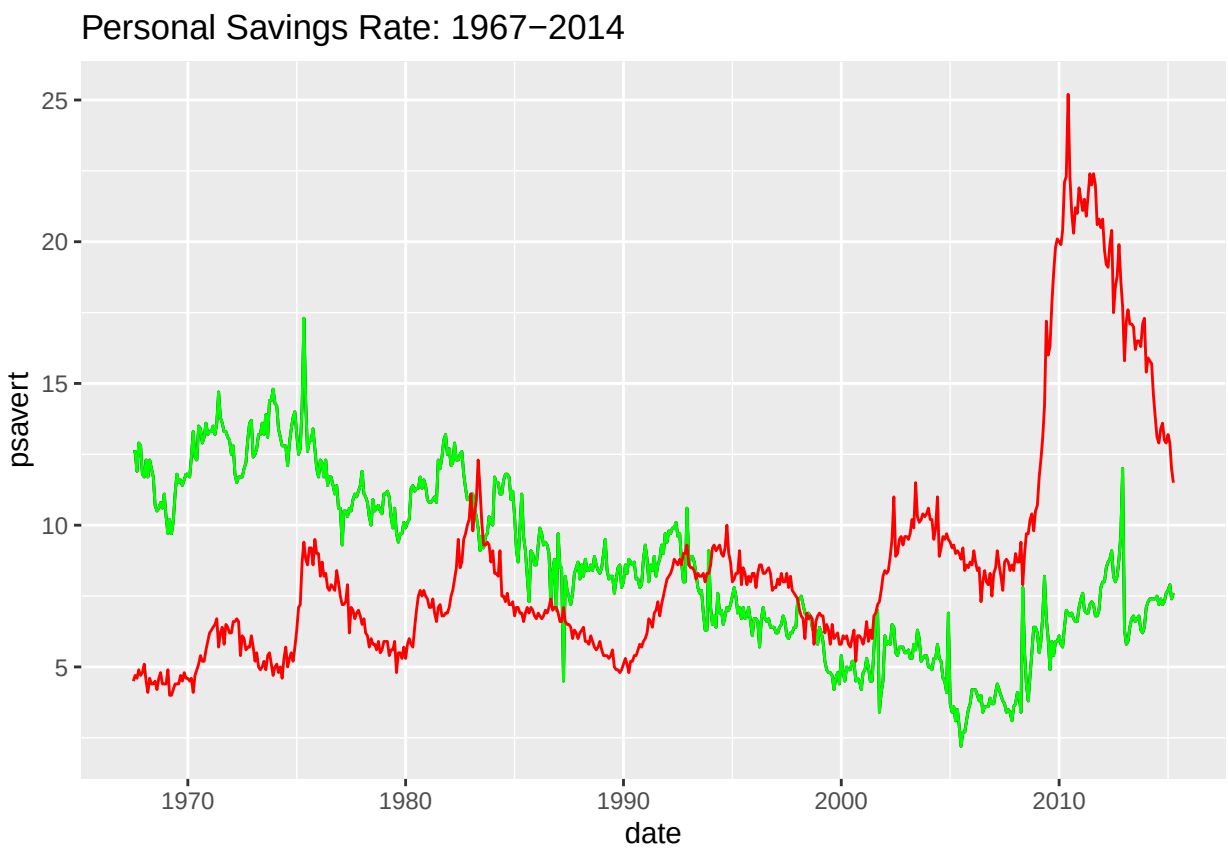
6. Add a title to the plot with the **ggtitle("Put title here")** sub-command. The title **"Personal Savings Rate: 1967-2014"** would be a good choice.

```r
myPlot <- myPlot + ggtitle("Personal Savings Rate: 1967-2014")
myPlot
```



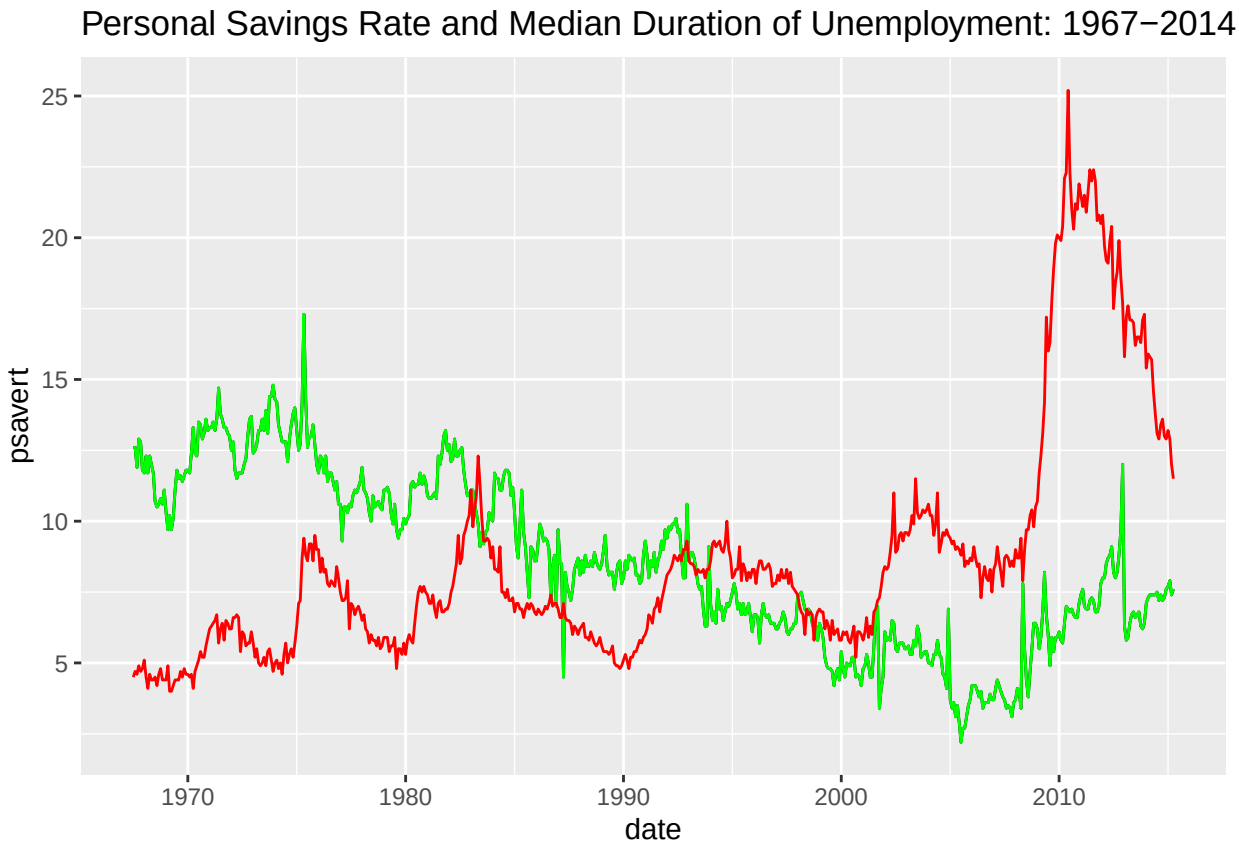
7. Add another data series to your plot to show the variable **uempmed** as a red line.

```
myPlot <- myPlot + geom_line(aes(y=uempmed), colour = "red")
myPlot
```



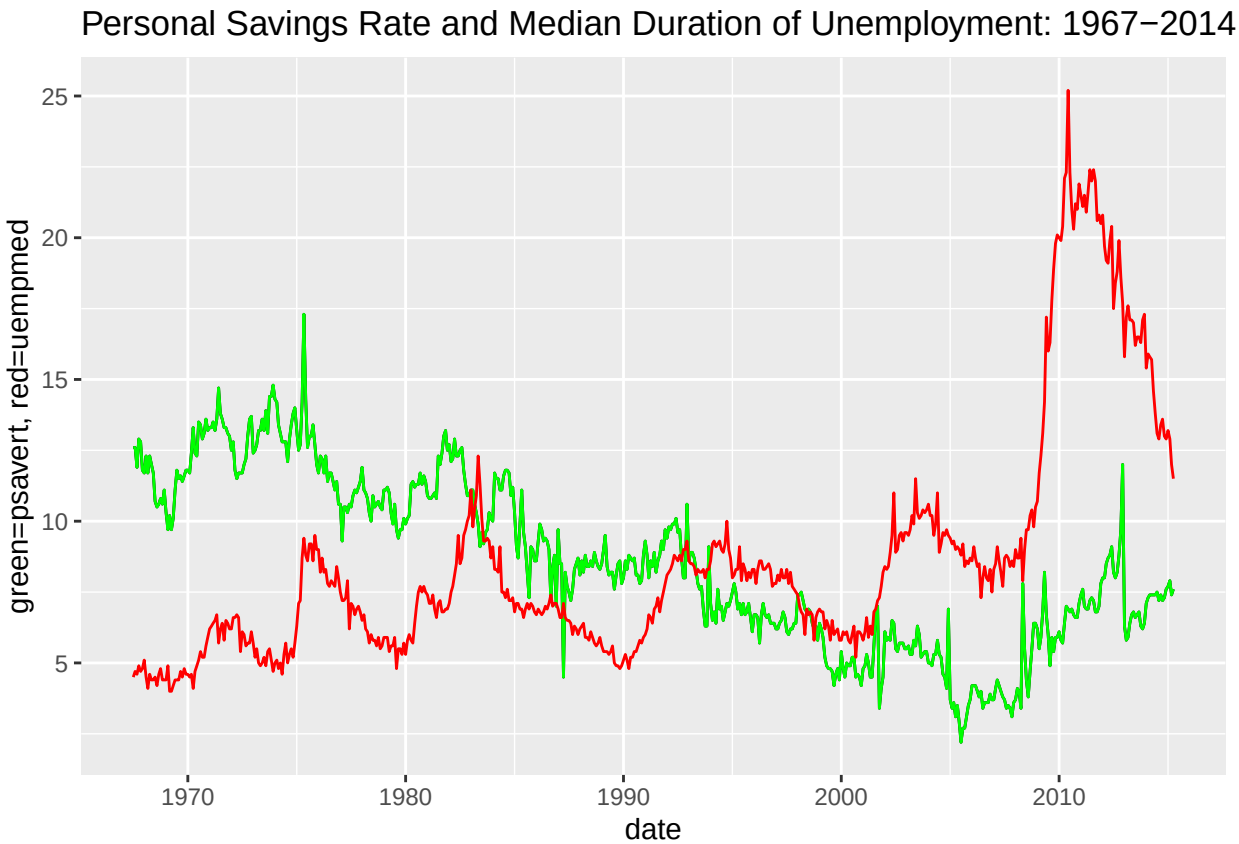
8. Change the title of the plot to mention both variables.

```
myPlot <- myPlot +
  ggtitle("Personal Savings Rate and Median Duration of Unemployment: 1967-2014")
myPlot
```



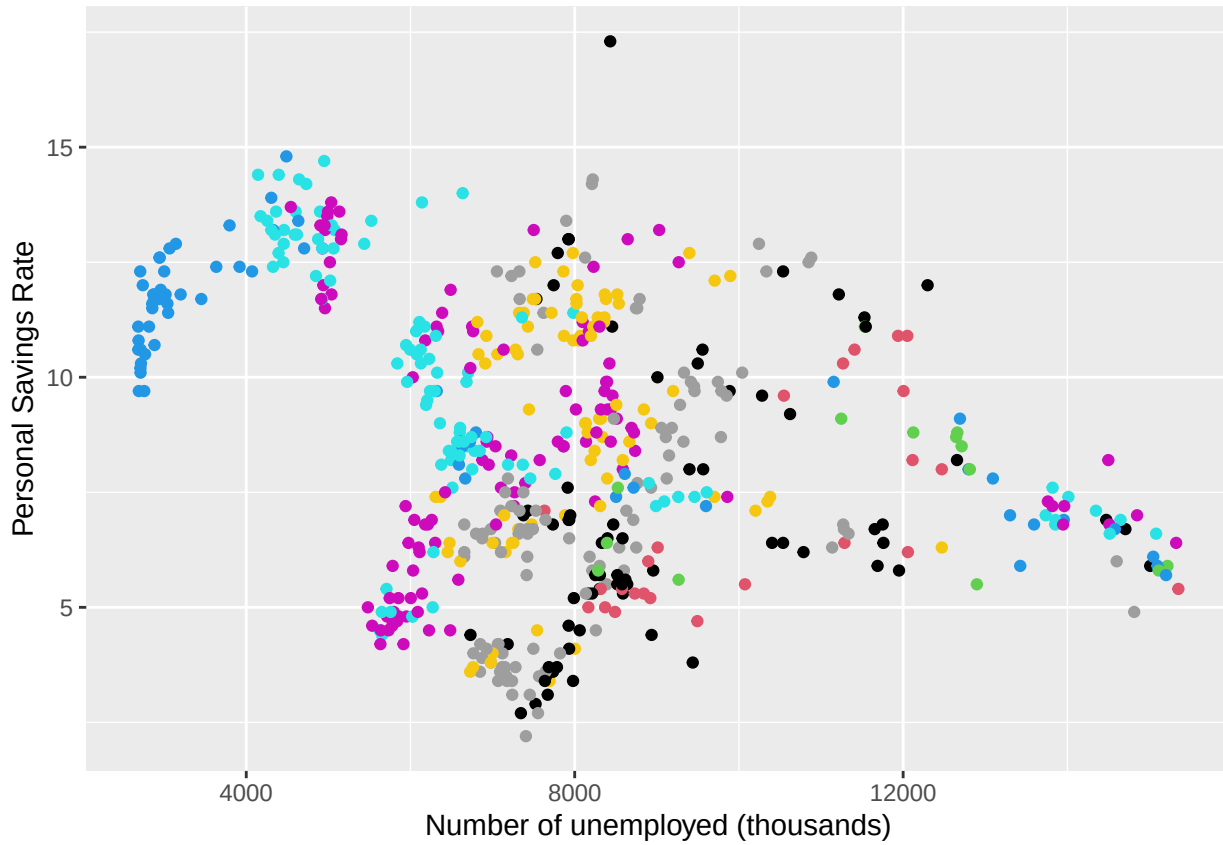
9. You can modify the axis labels in a ggplot with **ylab( )** and **xlab( )** subcommands. Change the axis labeling as needed to account for plotting both **psavert** and **uempmed** in the same window.

```
myPlot <- myPlot + ylab("green=psavert, red=uempmed")
myPlot
```



10. Create one last plot, creating a scatter plot, having the **unemploy** on the x-axis, **psavert** on the yaxis. Color each point based on the **uempmed**.

```
scatterPlot <- ggplot(economics, aes(x=unemploy, y=psavert)) + geom_point(colour = economics$uempmed) +
  xlab("Number of unemployed (thousands)") + ylab("Personal Savings Rate")
scatterPlot
```



11. Interpret (using comments in R) what you see in this last graph.

```
#There is a moderate negative correlation between personal savings rate and number of unemployed
#There is no correlation with the median duration of unemployment
```